

ACML100

Lecture 9

Determining Mechanism from Rate Law

- If the rate law is $r = k[A]^\alpha [B]^\beta \dots\dots [L]^\lambda$, the total composition of the reactants in the rate limiting step is $\alpha A + \beta B + \dots\dots + n H_2O$
- If the rate law is $r = k[A]^\alpha [B]^\beta \dots\dots [L]^\lambda / [M]^\mu [N]^\nu$, the total composition of the reactants in the rate limiting step is $\alpha A + \beta B + \dots\dots - \mu M - \nu N \dots\dots + n H_2O$
- A rate law, to be properly interpreted according to rule 1, must be written in terms of the predominate species in the reaction medium.

Mechanism

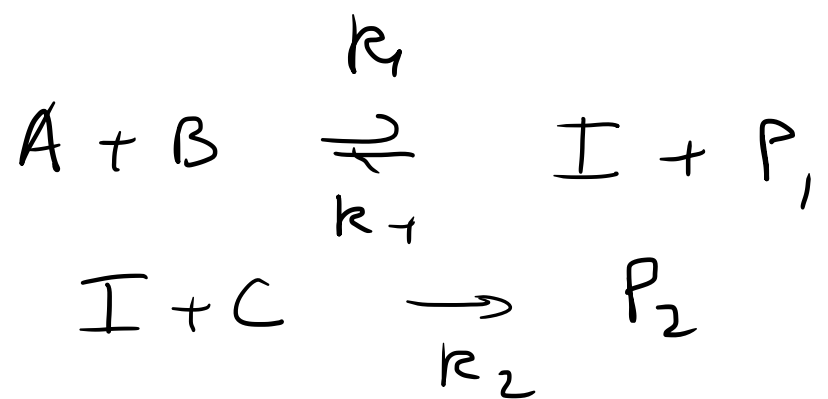
- The concentration dependence in rate law establish the elemental composition of Transition state and its charge.

Determining Mechanism from Rate Law

- A summation of n terms in the denominator implies a succession of n steps, where at least one step is reversible.
- Species whose concentrations appear in single-term denominators are produced in the step prior to rate controlling step.
- If there are three terms in the denominator, intermediate can disappear in three or more steps.

Determining Mechanism from Rate Law

- The number of positive terms in the rate law is the number of independent, parallel pathways. Negative terms represent the reverse reaction.
- Adding up the steps in a mechanism must yield the net chemical reaction; rapid reaction may follow the rate-controlling step.
- Alternative mechanisms leading to same pattern of activated complexes are not kinetically distinguishable.
- The order increases with increasing concentration when the reaction proceeds by parallel pathways but decreases when a series of steps occur.

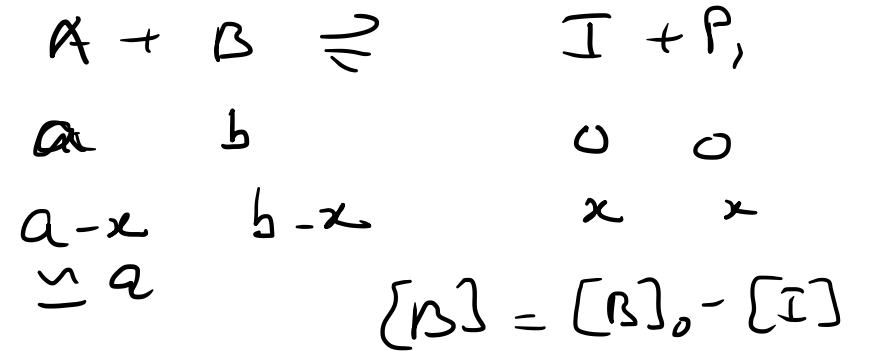


$$\text{Rate of } R_{\text{net}} = \frac{k_1 [A] [B] [C]}{k_{-1} [P_1] + k_2 [C]}$$

$$\frac{d[I]}{dt} = k_1 [A] [B] - k_{-1} [I] [P_1] - k_2 [I] [C] \stackrel{!}{=} 0$$

$$[I] = \frac{k_1 [A] [B]}{k_{-1} [P_1] + k_2 [C]}$$

$$\text{Rate of } R_{\text{net}} = \frac{k_1 k_2 [A] [B] [C]}{k_{-1} [P_1] + k_2 [C]}$$



$$\frac{d[I]}{dt} = k_1 [A][B] - k_{-1} [I][P_1] - k_2 [I][C] = 0$$

$$k_1 [A]_0 [B] - k_{-1} [I][P_1] - k_2 [I][C] = 0$$

$$k_1 [A]_0 \{ [B]_0 - [I] \} - k_{-1} [I][P_1] - k_2 [I][C] = 0$$

$$k_1 [A]_0 [B]_0 - [I] \{ k_1 [A]_0 + k_{-1} [P_1] + k_2 [C] \} = 0$$

$$[I] = \frac{k_1 [A]_0 [B]_0}{k_1 [A]_0 + k_{-1} [P_1] + k_2 [C]_0}$$